

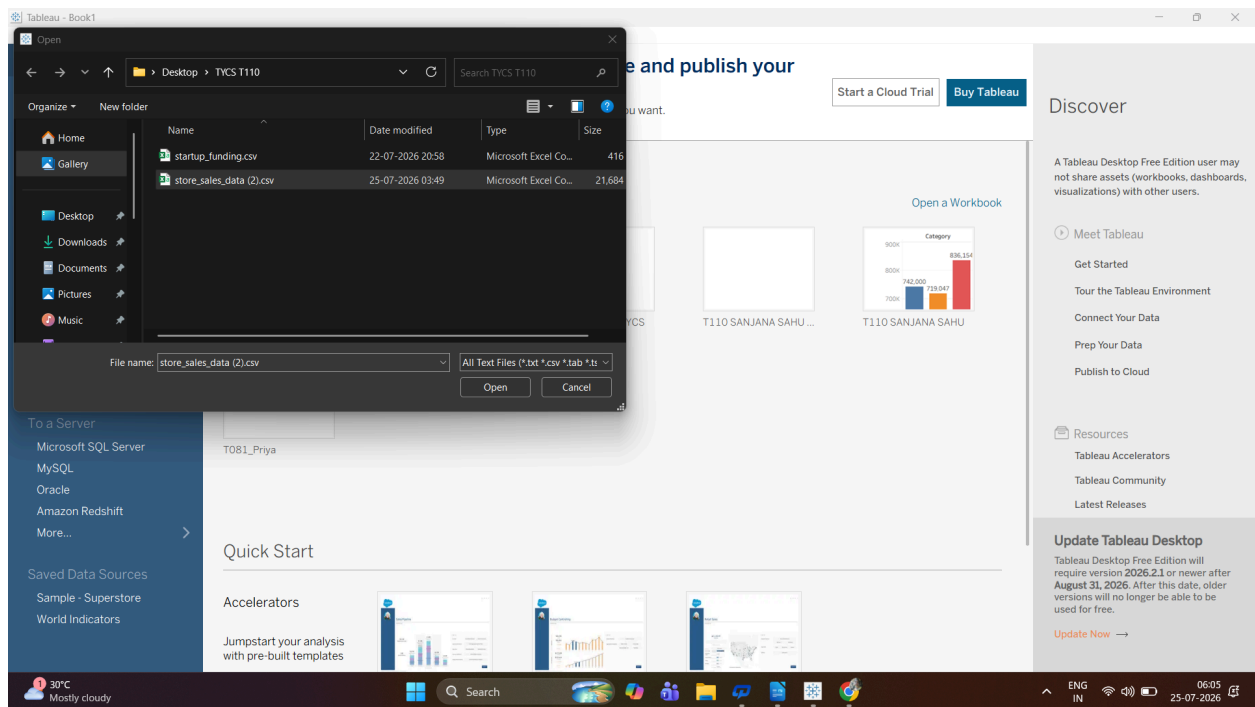
Data Visualization with Power BI and Tableau

Lesson 4: Performing Data Analysis using Calculations in Tableau

- a) Create basic calculated fields using formulas
- b) Perform row-level calculations
- c) Perform aggregate-level calculations
- d) Implement FIXED Level of Detail (LOD) calculations
- e) Use INCLUDE and EXCLUDE LOD expressions
- f) Create and use parameters for dynamic analysis
- g) Apply calculations to fix data issues and enhance insights

A: Creating Basic Calculated Fields Using Formulas

1. Connecting to the Dataset



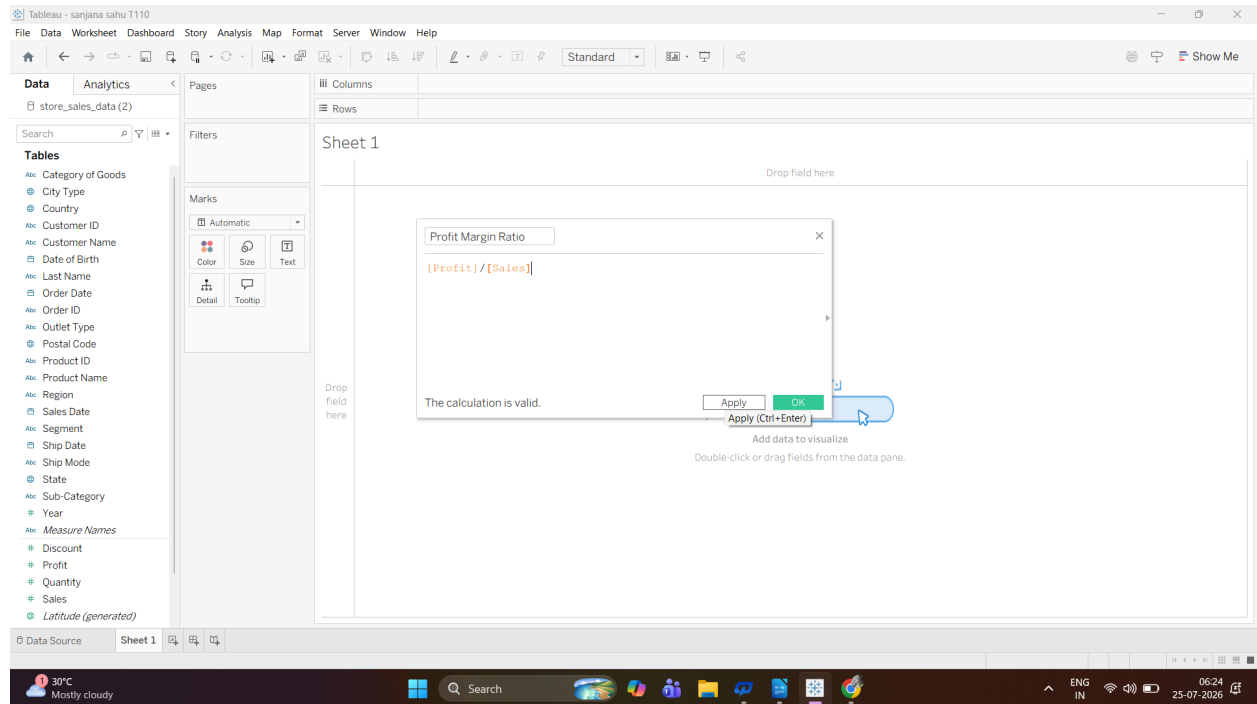
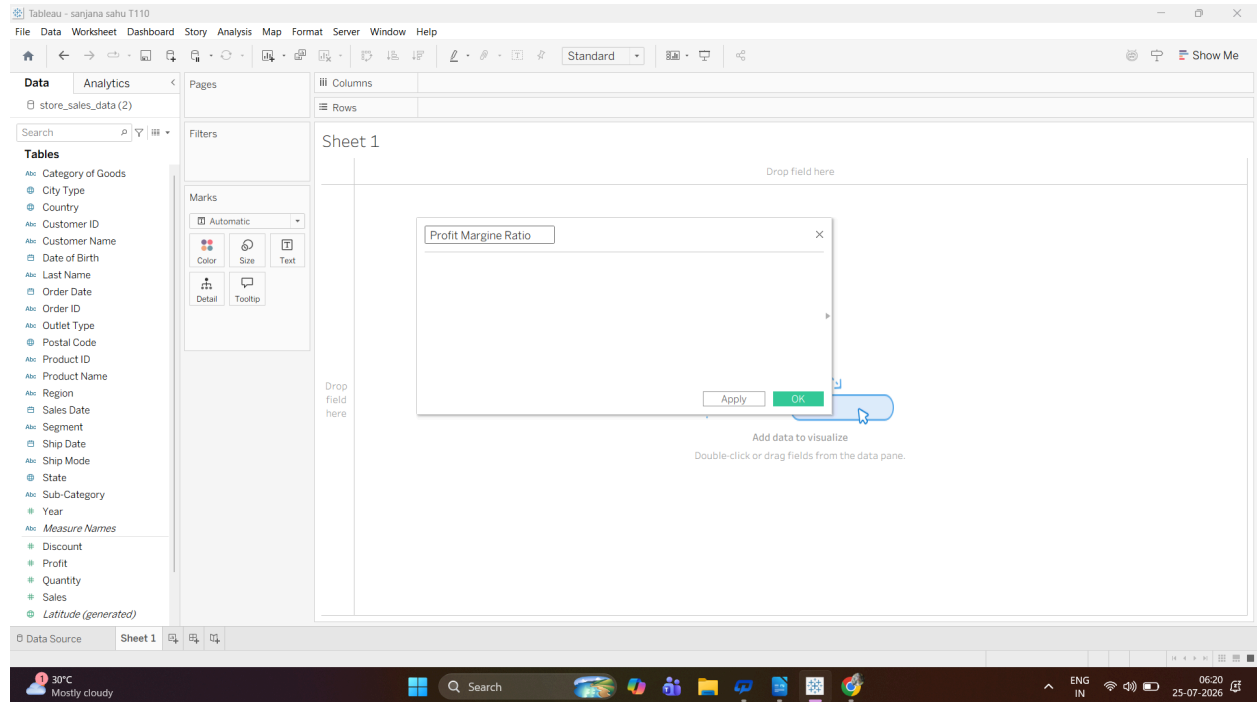


Tableau - sanjana.sahu T110

File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

store_sales_data (2)

Search

Tables

- Category of Goods
- City Type
- Country
- Customer ID
- Customer Name
- Date of Birth
- Last Name
- Order Date
- Order ID
- Outlet Type
- Postal Code
- Product ID
- Product Name
- Region
- Sales Date
- Segment
- Ship Date
- Ship Mode
- State
- Sub-Category
- Year
- Measure Names
- Discount
- Profit
- Profit Margin Ratio
- Quantity
- Sales

Columns

Rows

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Drop field here

Drop field here

Add data to visualize

Double-click or drag fields from the data pane.

Data Source Sheet 1

30°C Mostly cloudy

Search

ENG IN 06:25 25-07-2026

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File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

store_sales_data (2)

Search

Tables

- Category of Goods
- City Type
- Country
- Customer ID
- Customer Name
- Date of Birth
- Last Name
- Order Date
- Order ID
- Outlet Type
- Postal Code
- Product ID
- Product Name
- Region
- Sales Date
- Segment
- Ship Date
- Ship Mode
- State
- Sub-Category
- Year
- Measure Names
- Discount
- Profit
- Profit Margin Ratio
- Quantity
- Sales

Columns

Category of Goods

Rows

Filters

Marks

Automatic

Color Size Text

Detail Tooltip

Sheet 1

Category of Goods	
Dairy Products	Abc
Electric Appliances	Abc
Fast Food	Abc
Furniture	Abc
Household Items	Abc
Seasonal Fruits & V..	Abc

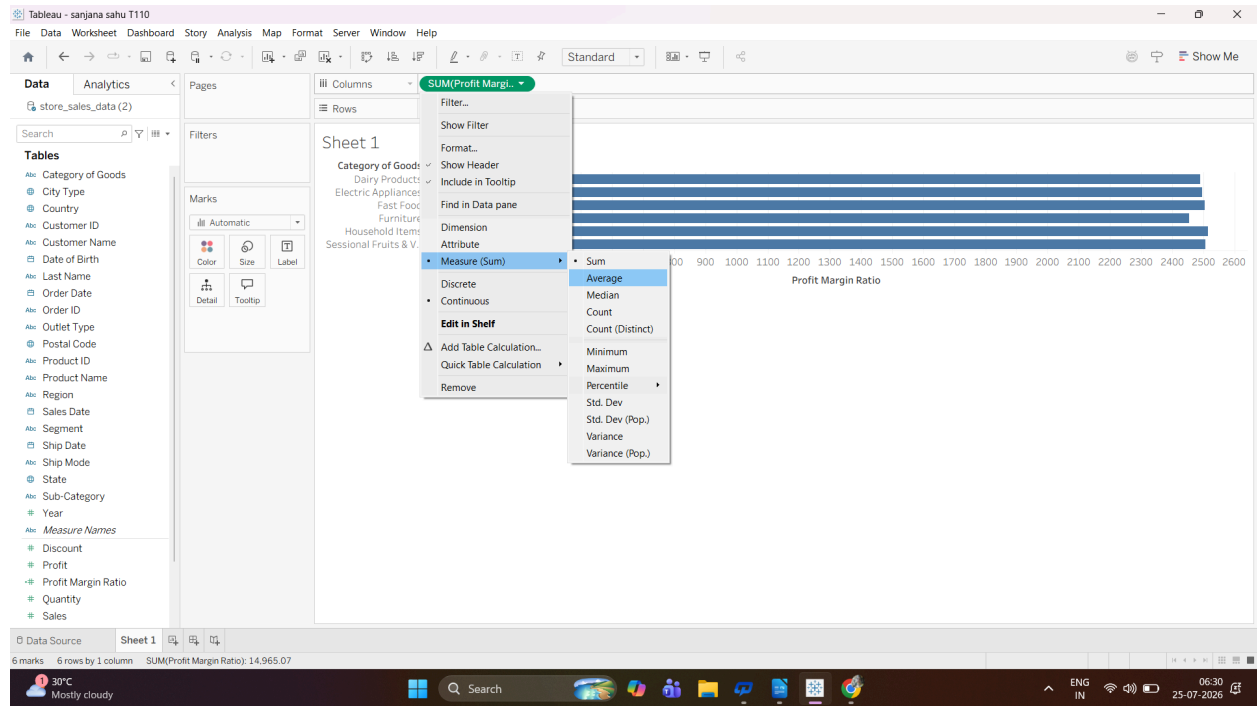
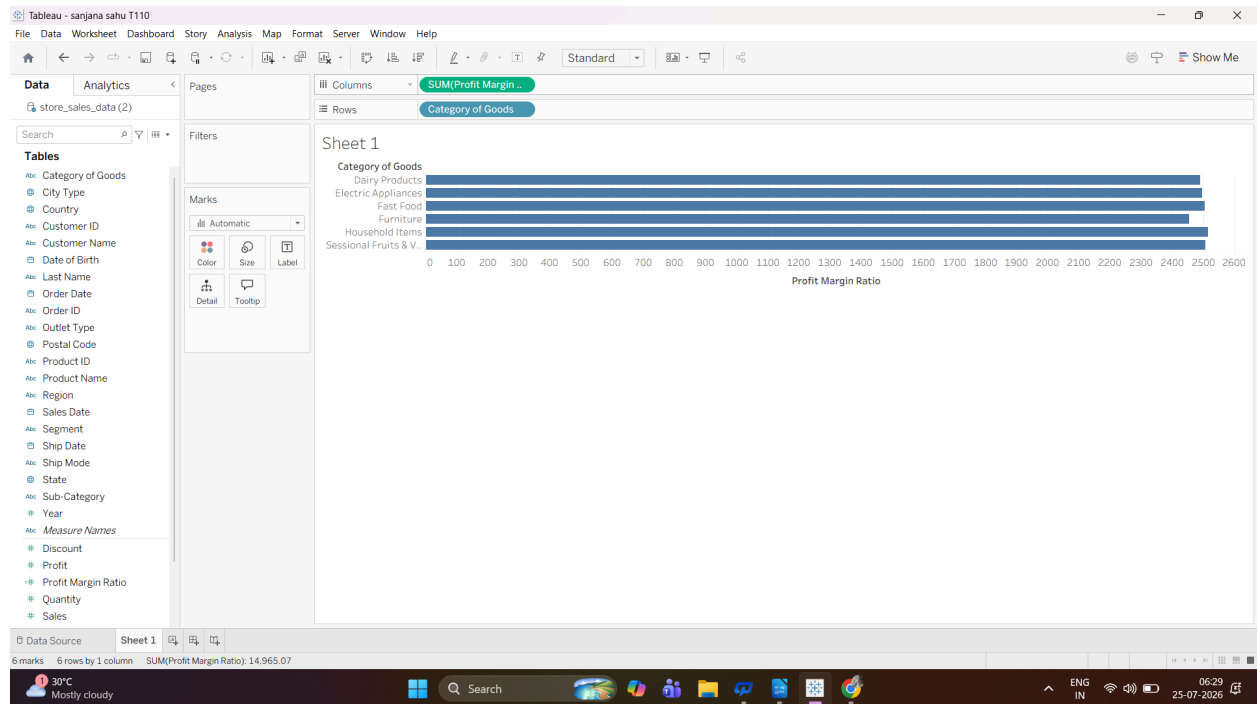
Data Source Sheet 1

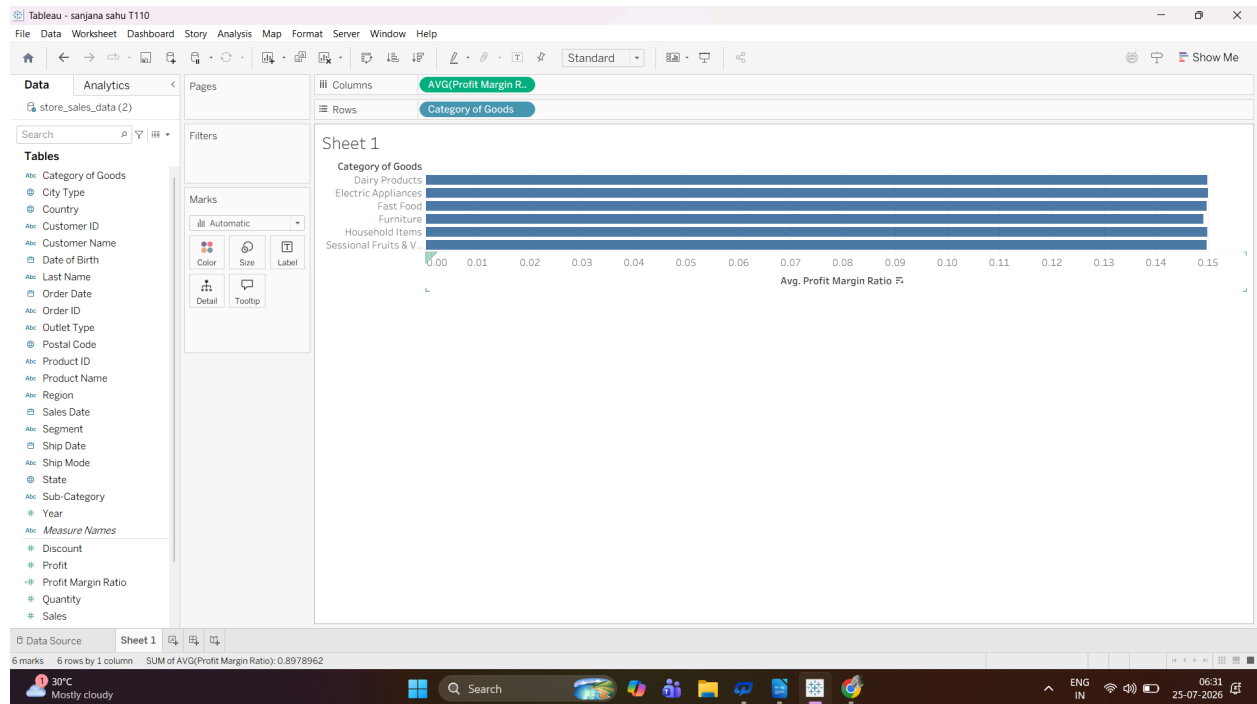
6 marks 6 rows by 1 column

30°C Mostly cloudy

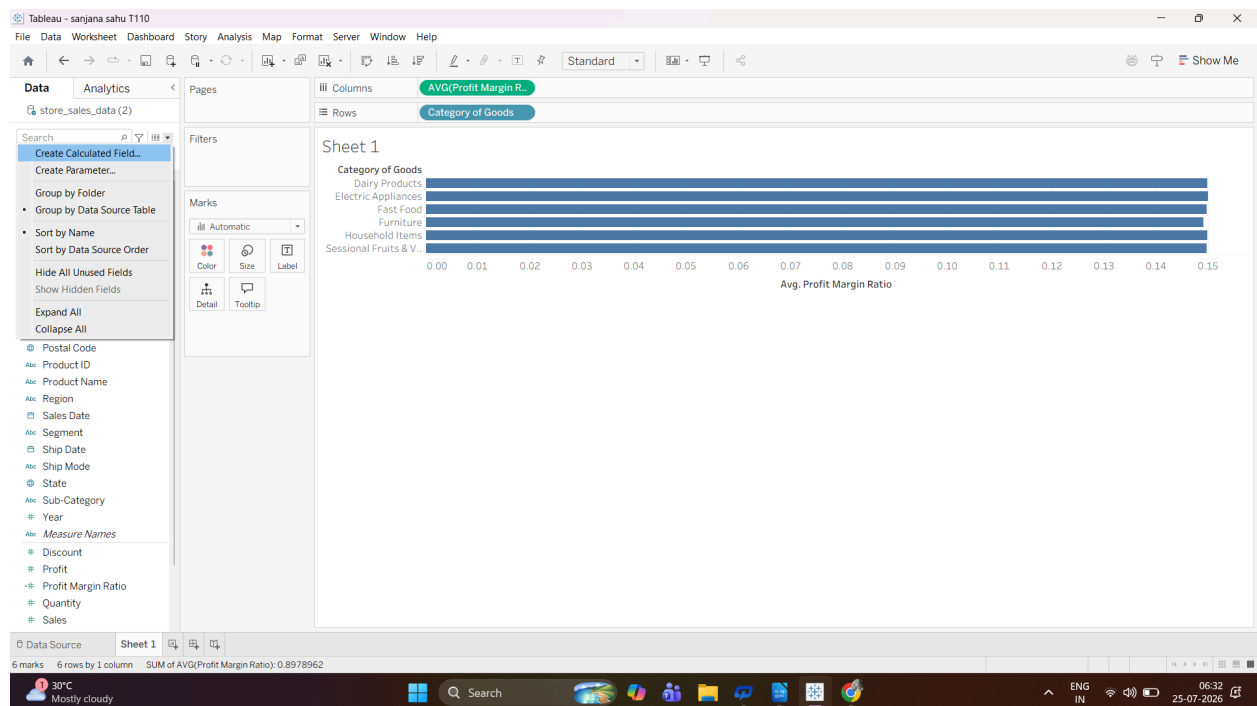
Search

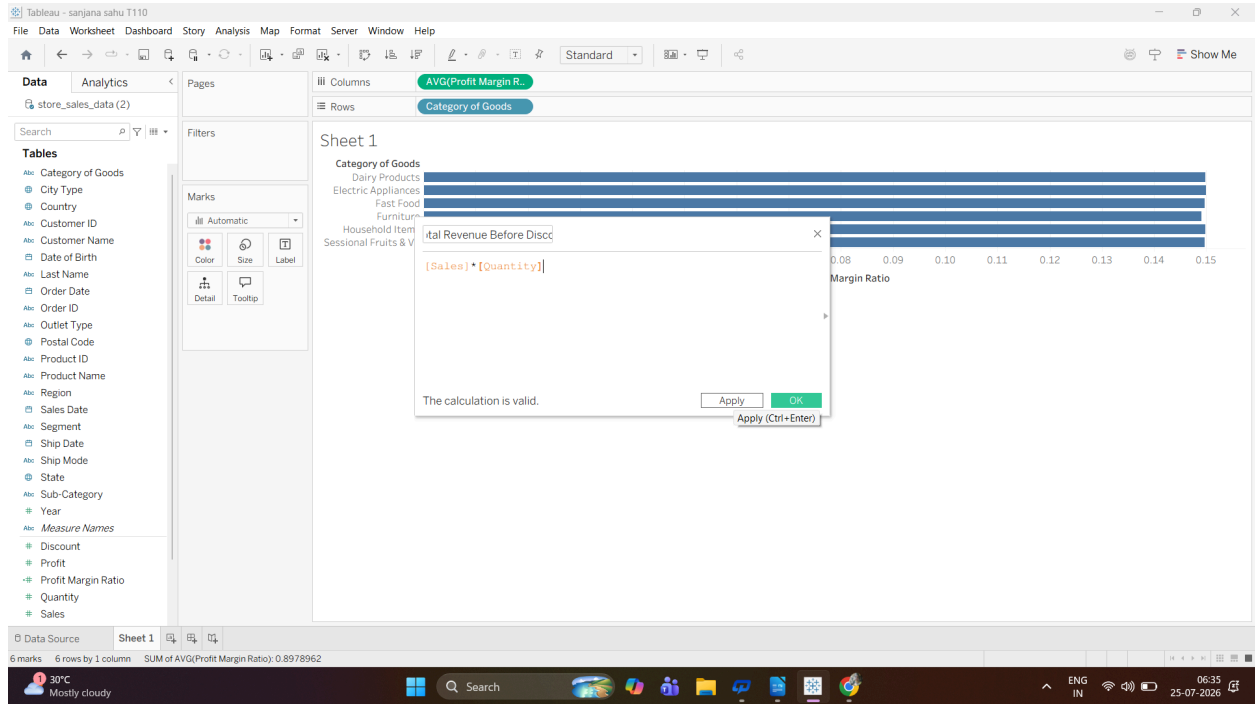
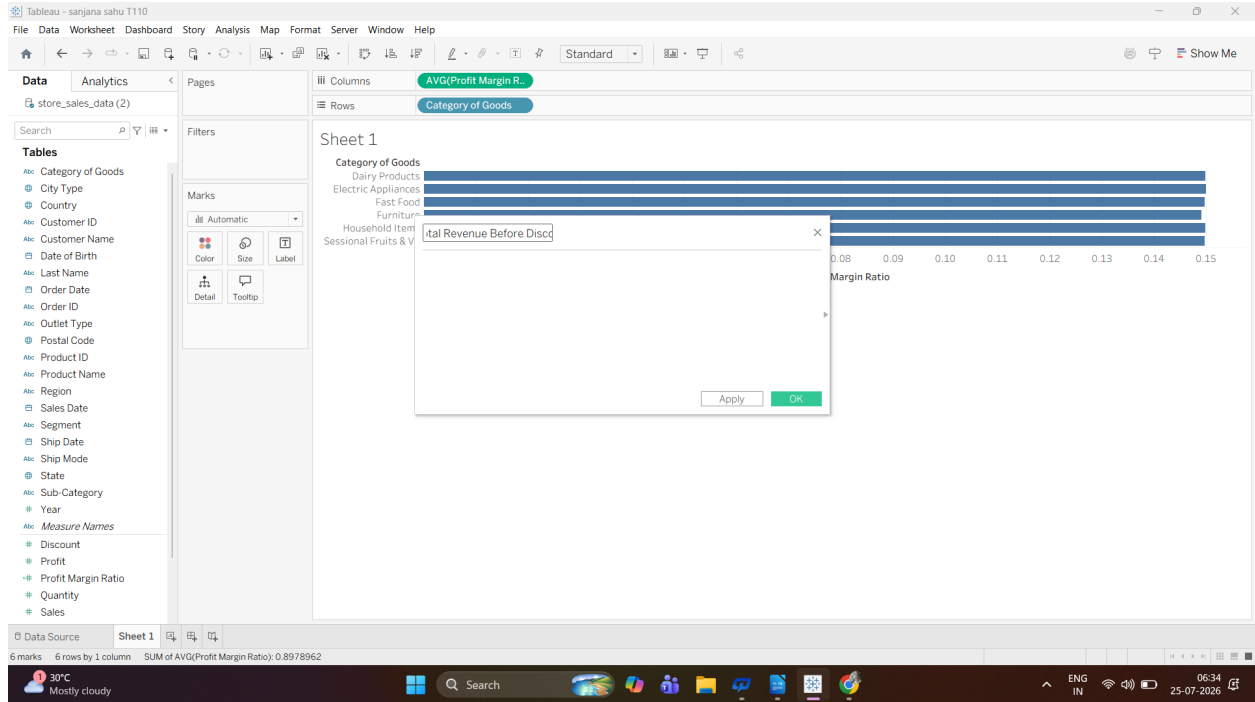
ENG IN 06:28 25-07-2026

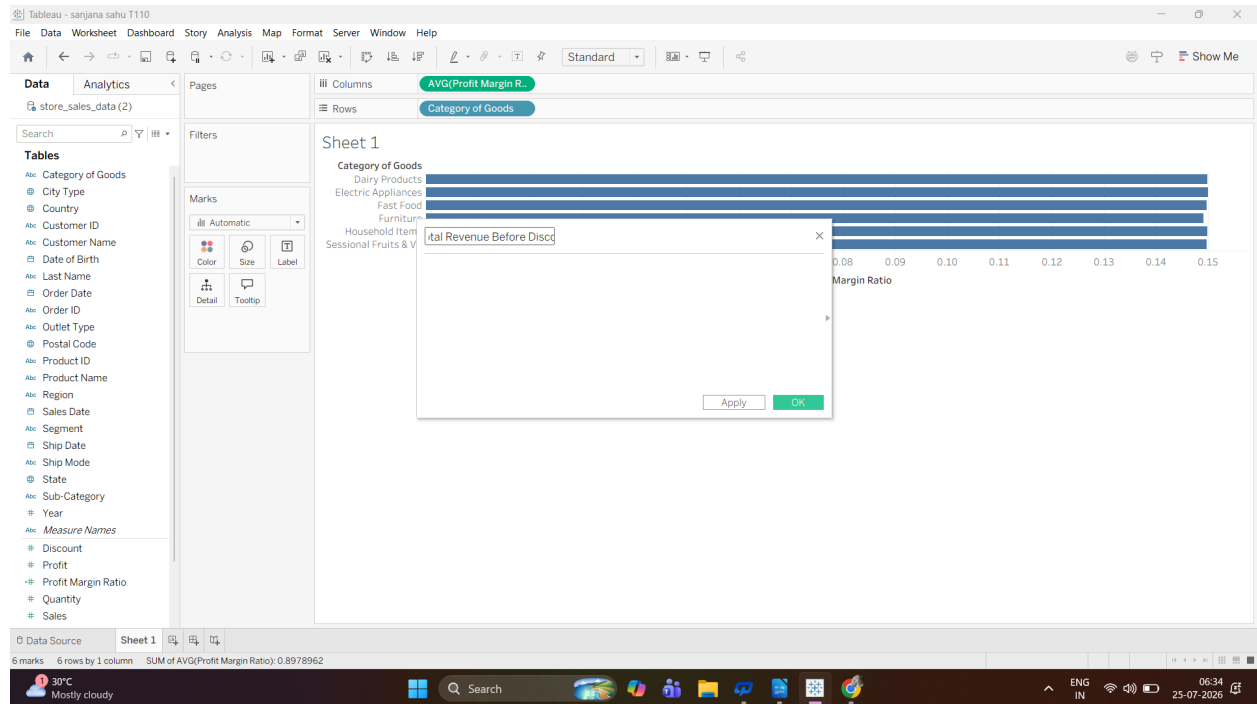
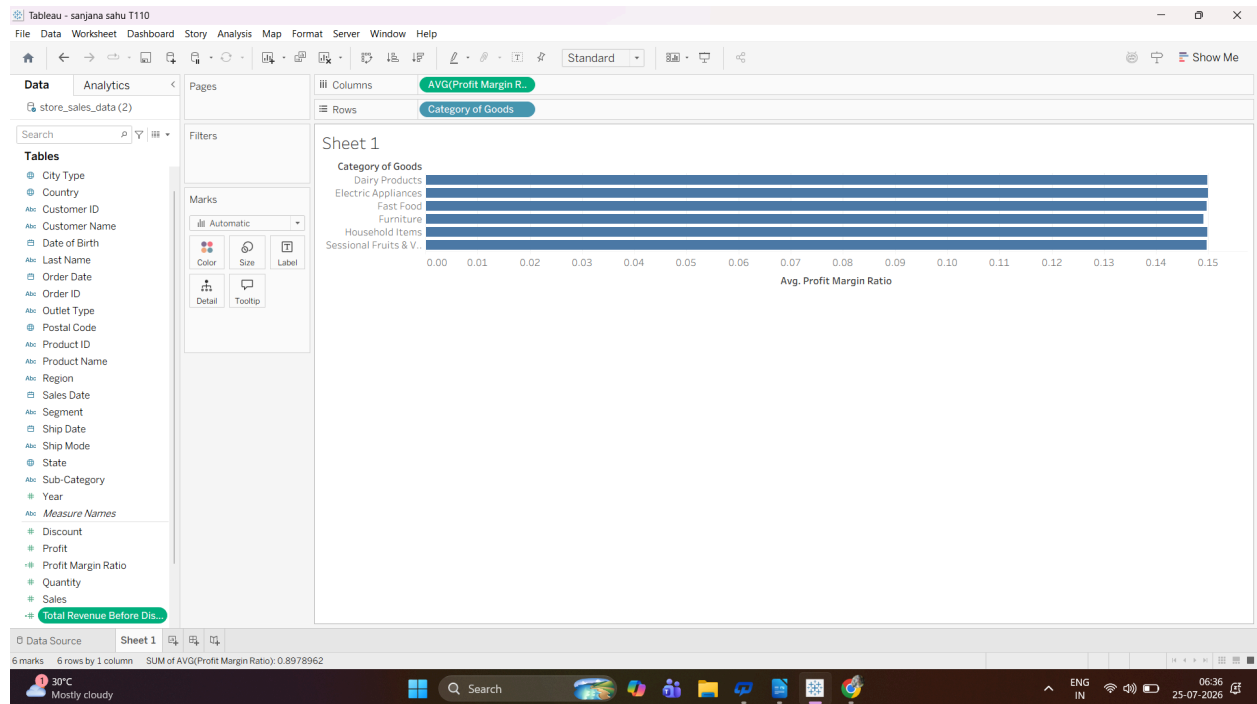


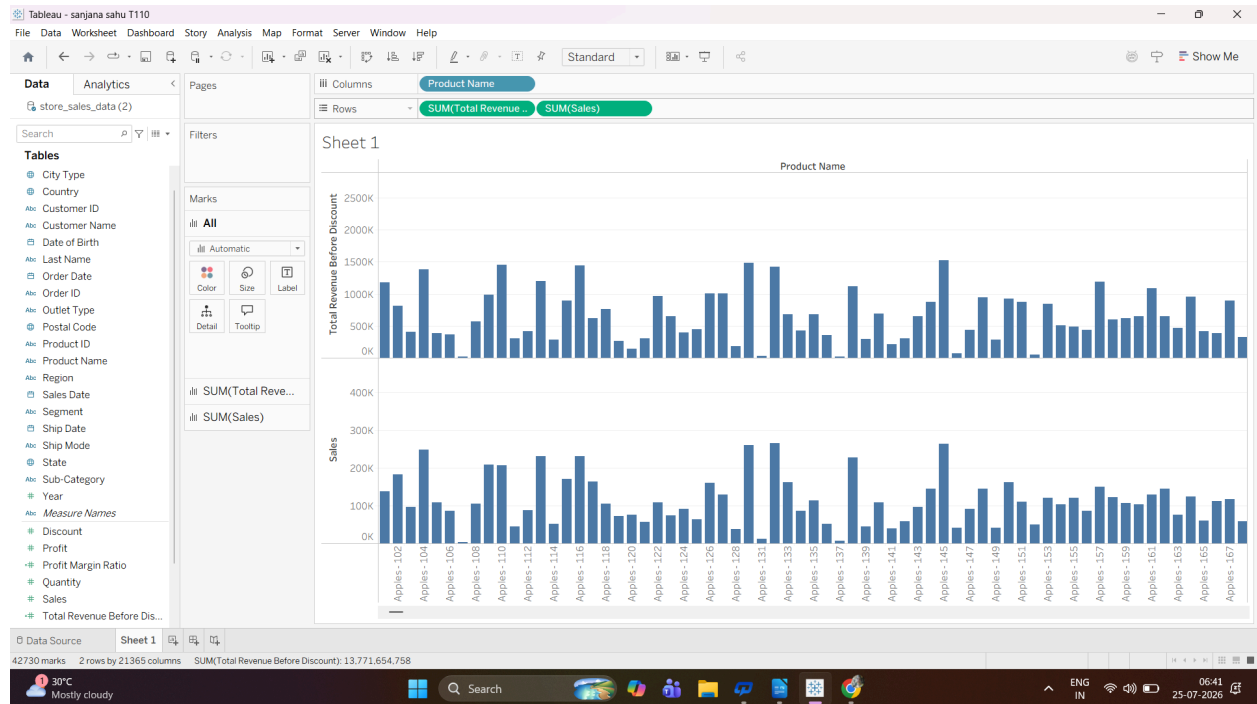
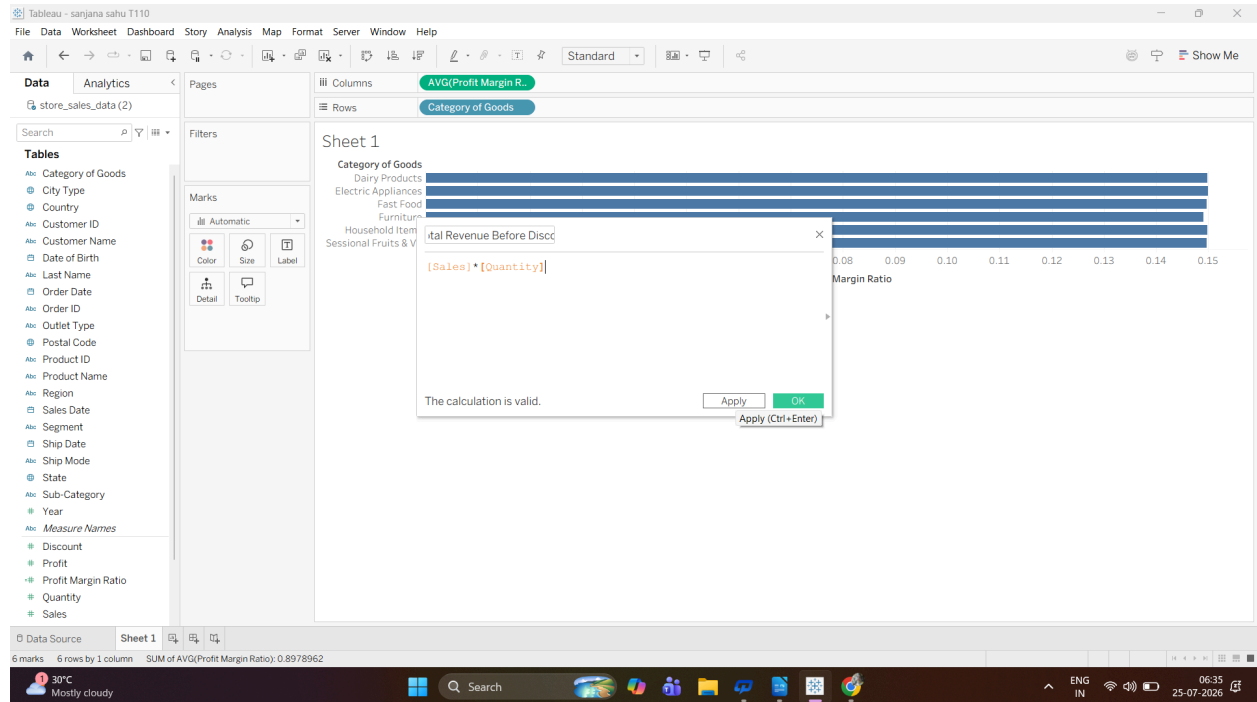


b) Perform row-level calculations

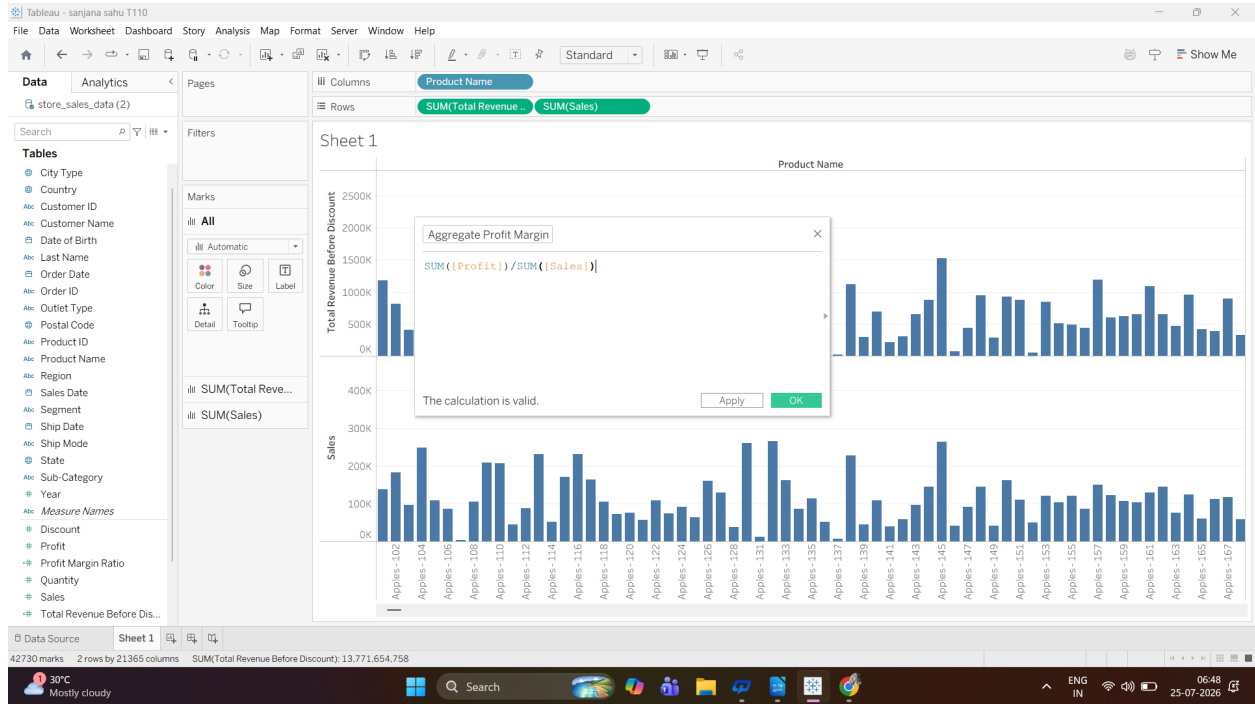
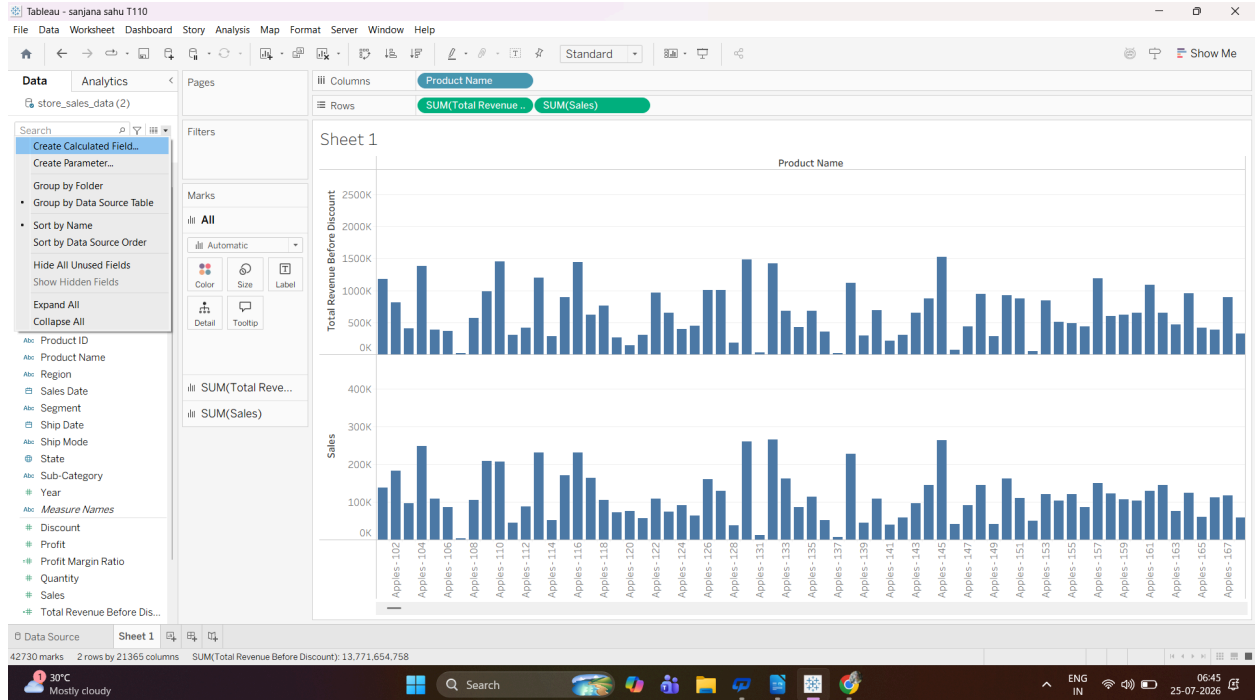


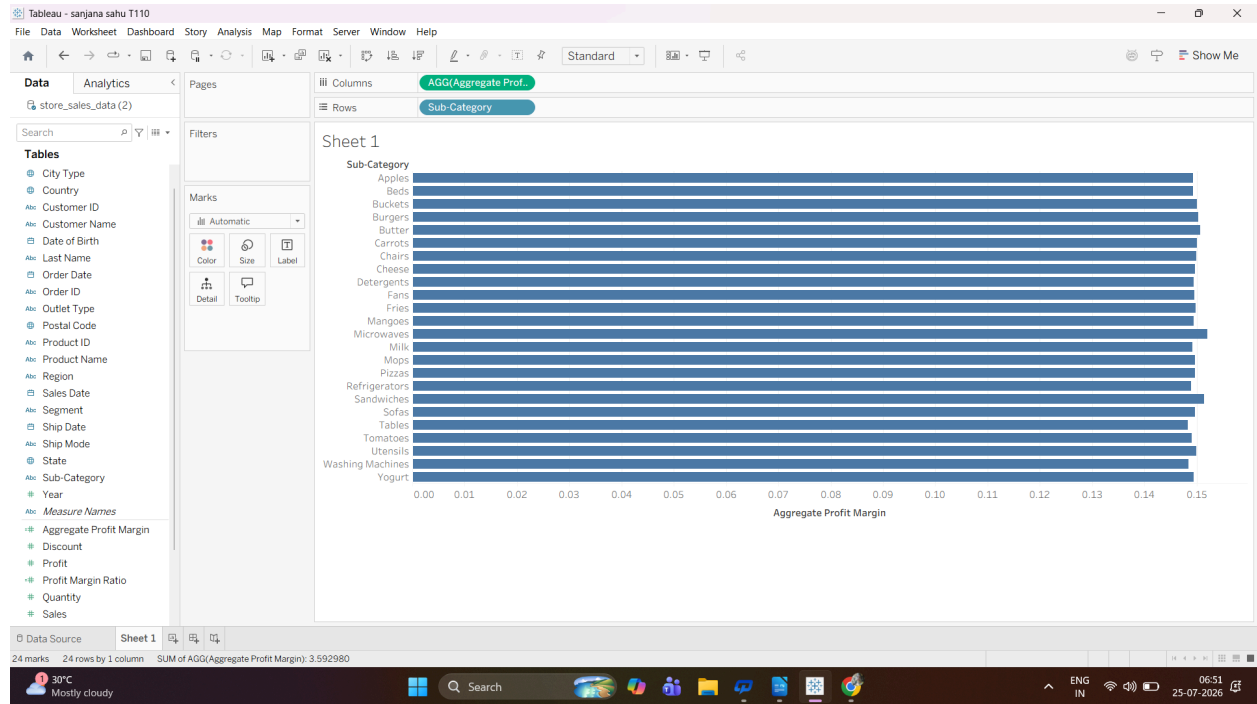




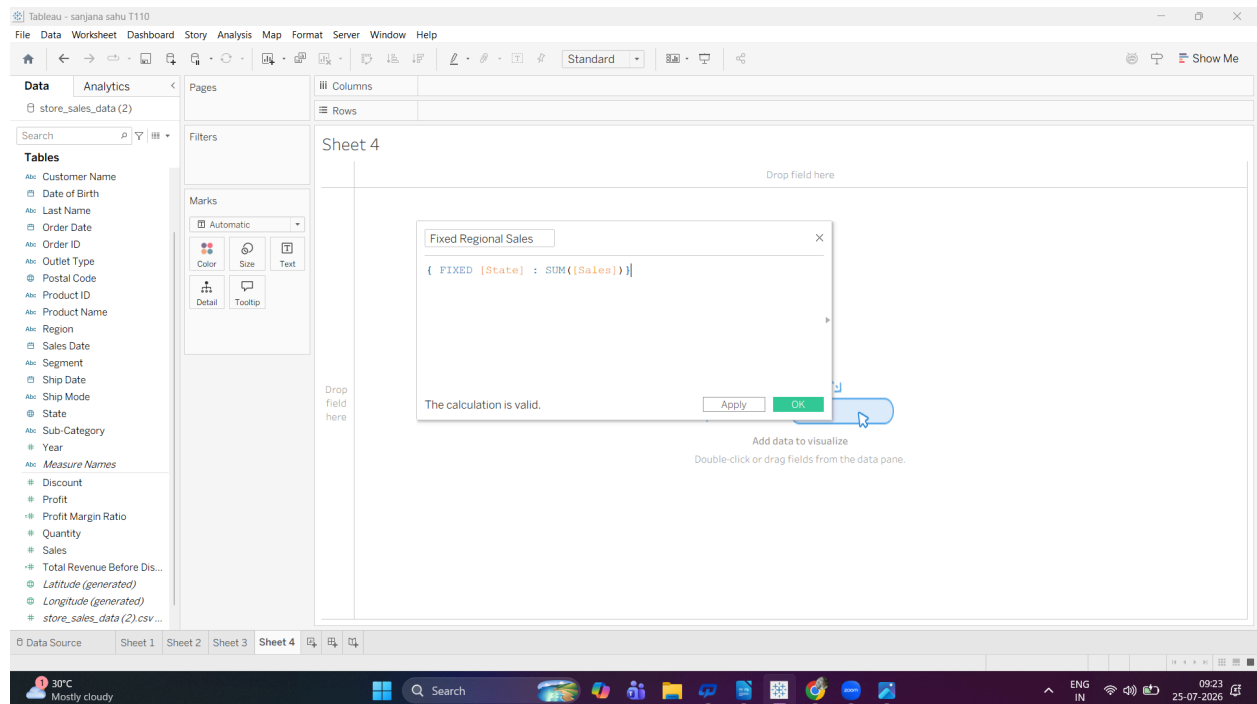


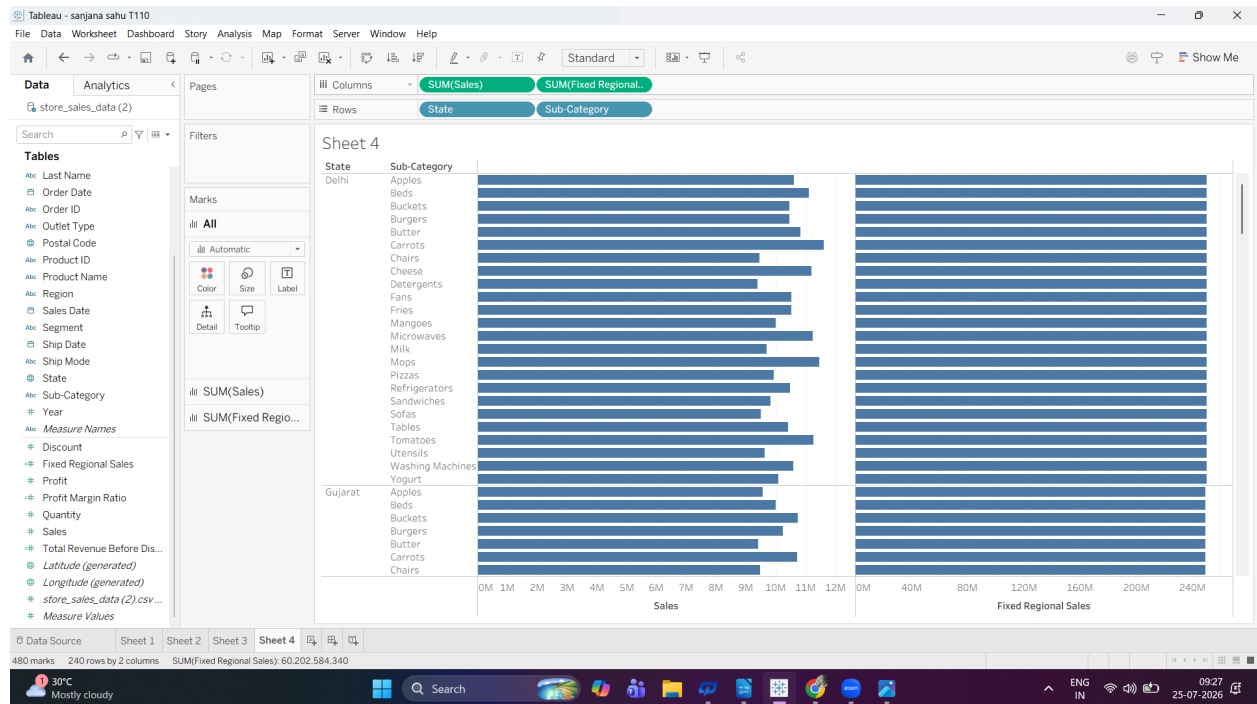
c) Perform aggregate-level calculations



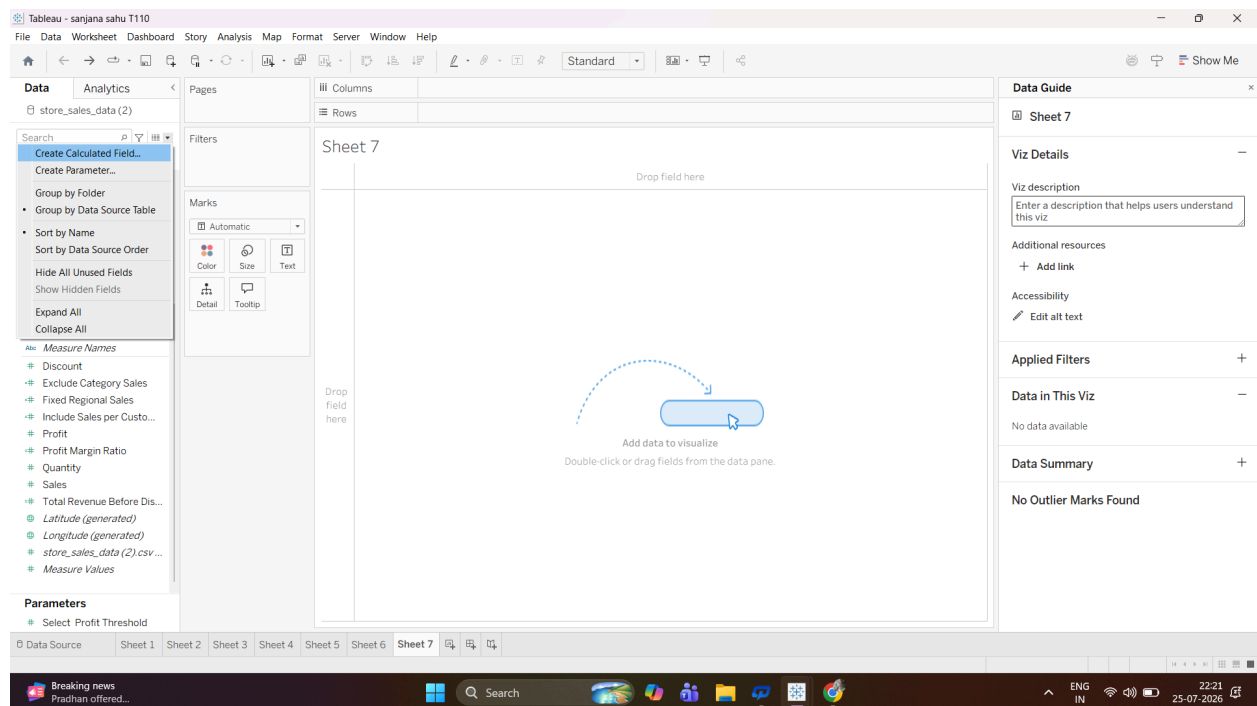


D]Implement FIXED Level of Detail (LOD) calculations





E]Use INCLUDE and EXCLUDE LOD expressions



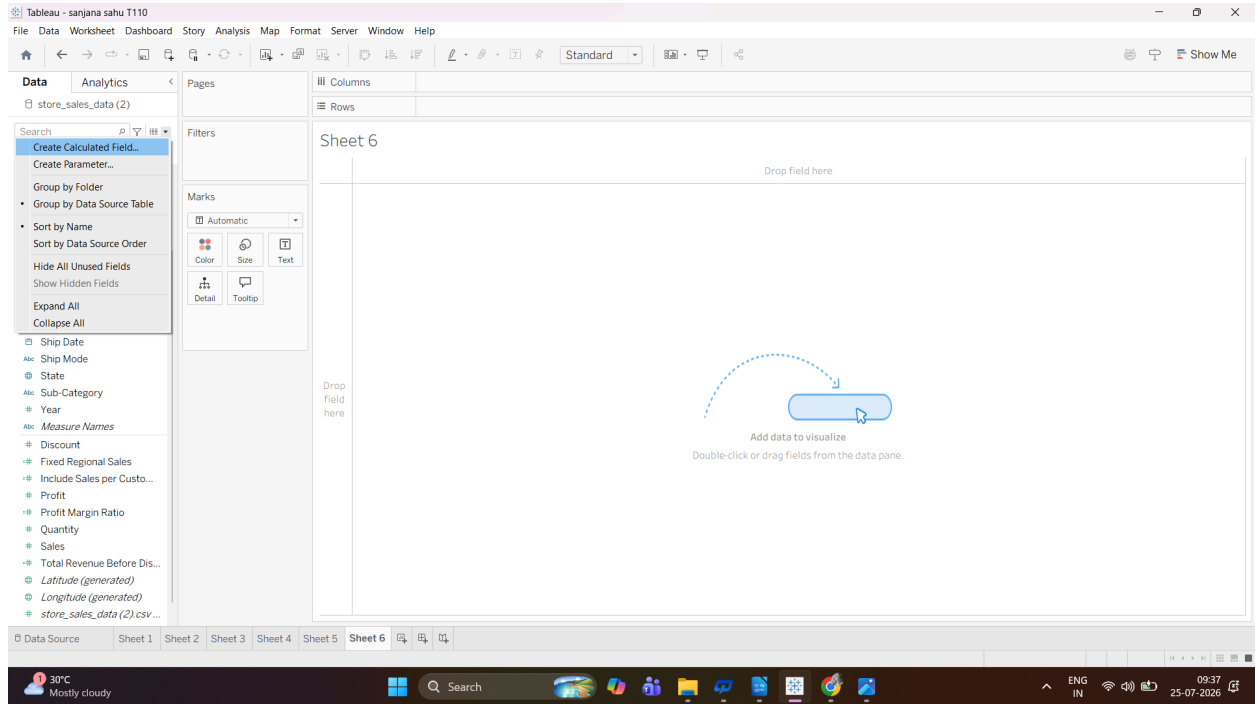
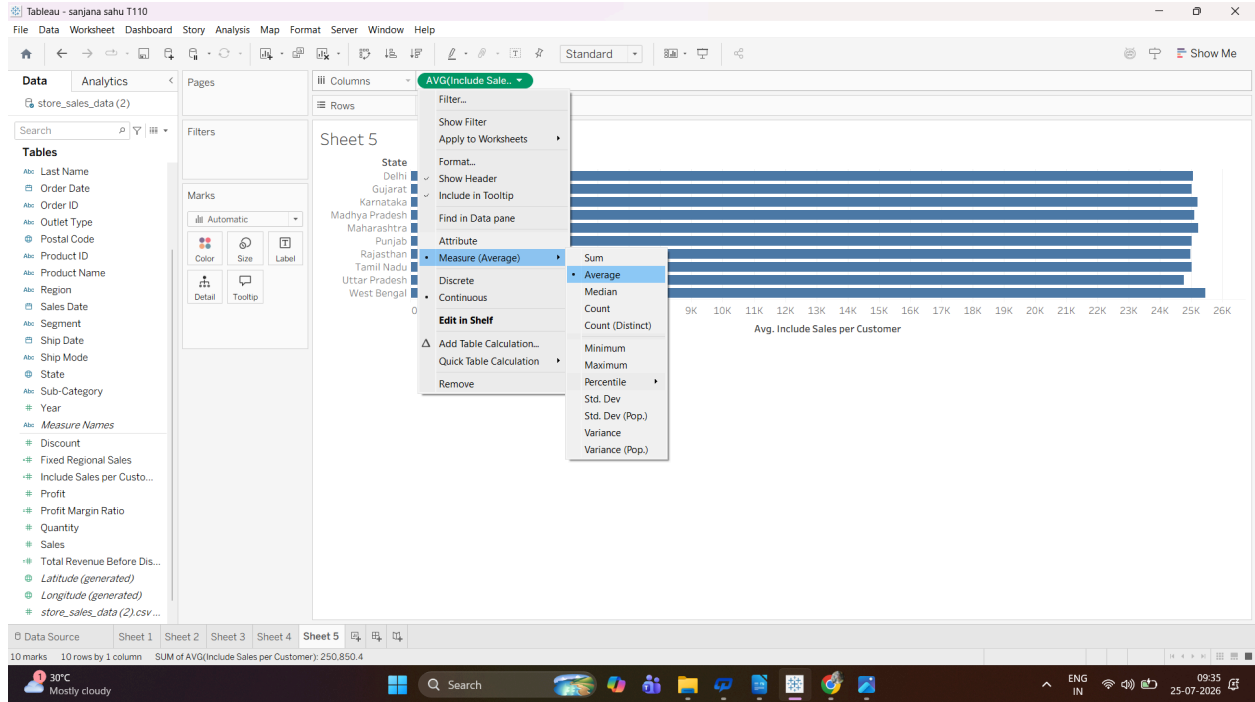


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File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Analytics

store_sales_data (2)

Search

Tables

- Abc Last Name
- Abc Order Date
- Abc Order ID
- Abc Outlet Type
- Abc Postal Code
- Abc Product ID
- Abc Product Name
- Abc Region
- Abc Sales Date
- Abc Segment
- Abc Ship Date
- Abc Ship Mode
- Abc State
- Abc Sub-Category
- Abc Year
- Abc Measure Names
- # Discount
- # Fixed Regional Sales
- # Include Sales per Custo...
- # Profit
- # Profit Margin Ratio
- # Quantity
- # Sales
- # Total Revenue Before Dis...
- # Latitude (generated)
- # Longitude (generated)
- # store_sales_data (2).csv...

Filters

Marks

Automatic

Color Size Text Detail Tooltip

Drop field here

Drop field here

Drop field here

Exclude Category Sales

{EXCLUDE [Sub-Category] : SUM([Sales])}

The calculation is valid.

Apply OK

Add data to visualize

Double-click or drag fields from the data pane.

Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6

30°C Mostly cloudy

Search

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File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Analytics

store_sales_data (2)

Search

Tables

- Abc Order ID
- Abc Outlet Type
- Abc Postal Code
- Abc Product ID
- Abc Product Name
- Abc Region
- Abc Sales Date
- Abc Segment
- Abc Ship Date
- Abc Ship Mode
- Abc State
- Abc Sub-Category
- Abc Year
- Abc Measure Names
- # Discount
- # Exclude Category Sales
- # Fixed Regional Sales
- # Include Sales per Custo...
- # Profit
- # Profit Margin Ratio
- # Quantity
- # Sales
- # Total Revenue Before Dis...
- # Latitude (generated)
- # Longitude (generated)
- # store_sales_data (2).csv...
- # Measure Values

Filters

Marks

Automatic

Color Size Label Detail Tooltip

Sub-Category ATTR(Exclude Cate...

Sheet 6

Sub-Cate...

Sub-Category	Exclude Category Sales	Sales
Apples	28	18
Apples	18	08
Beds	28	18
Beds	18	08
Buckets	28	18
Buckets	18	08
Burgers	28	18
Burgers	18	08
Butter	28	18
Butter	18	08
Carrots	28	18
Carrots	18	08
Chairs	28	18

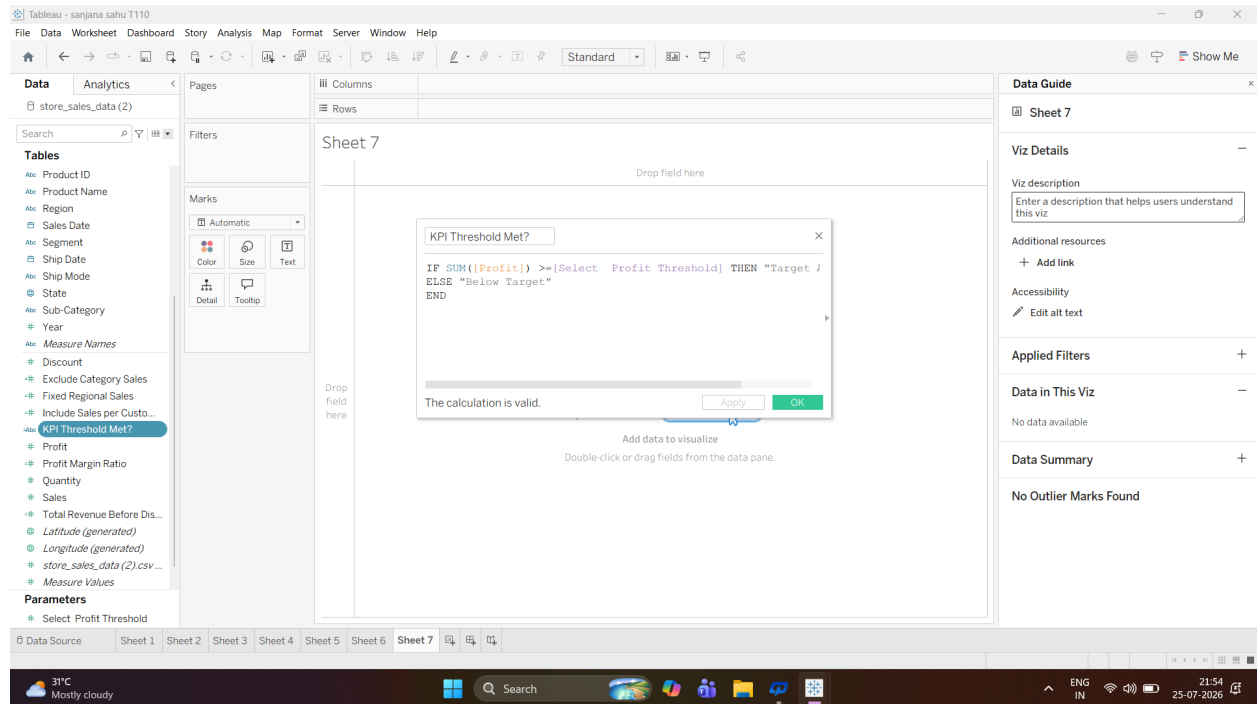
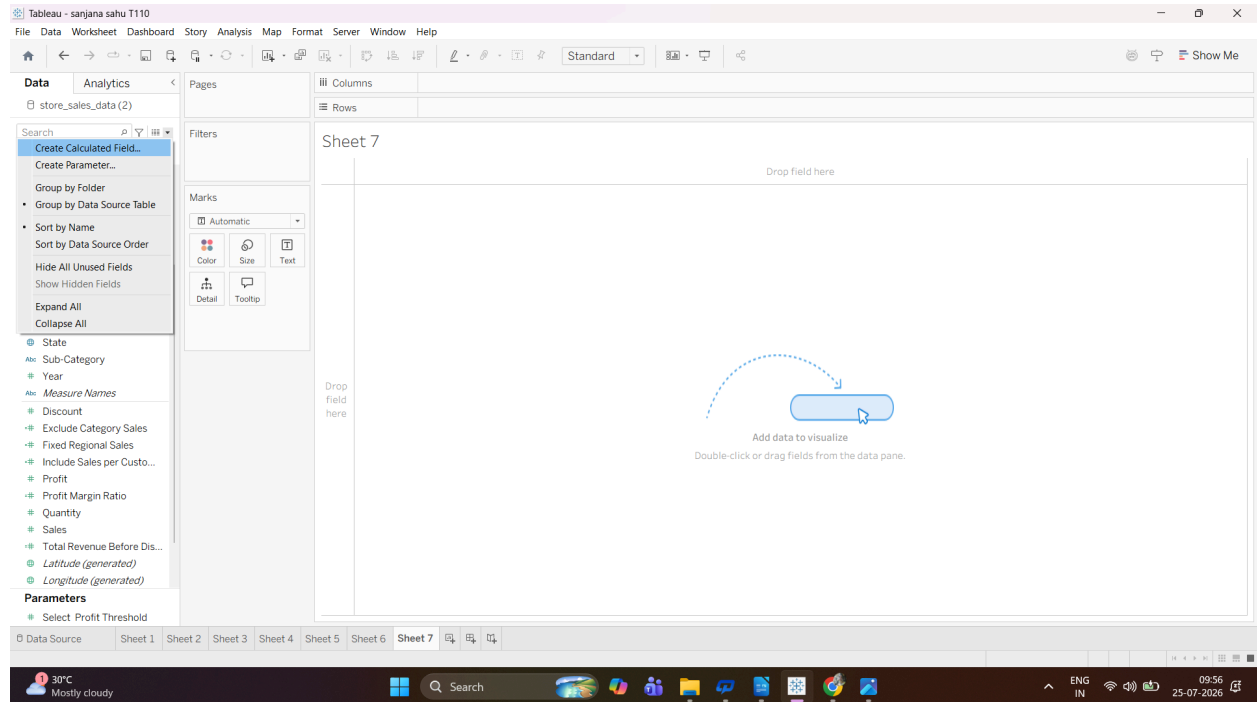
Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6

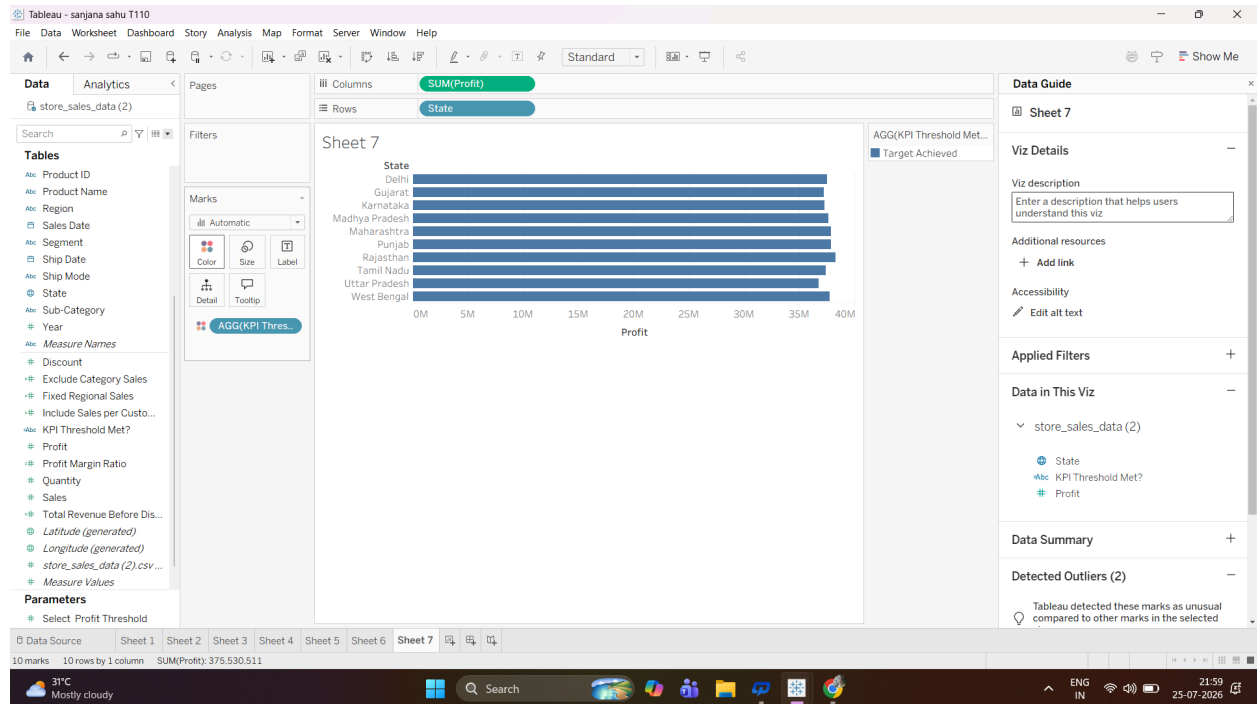
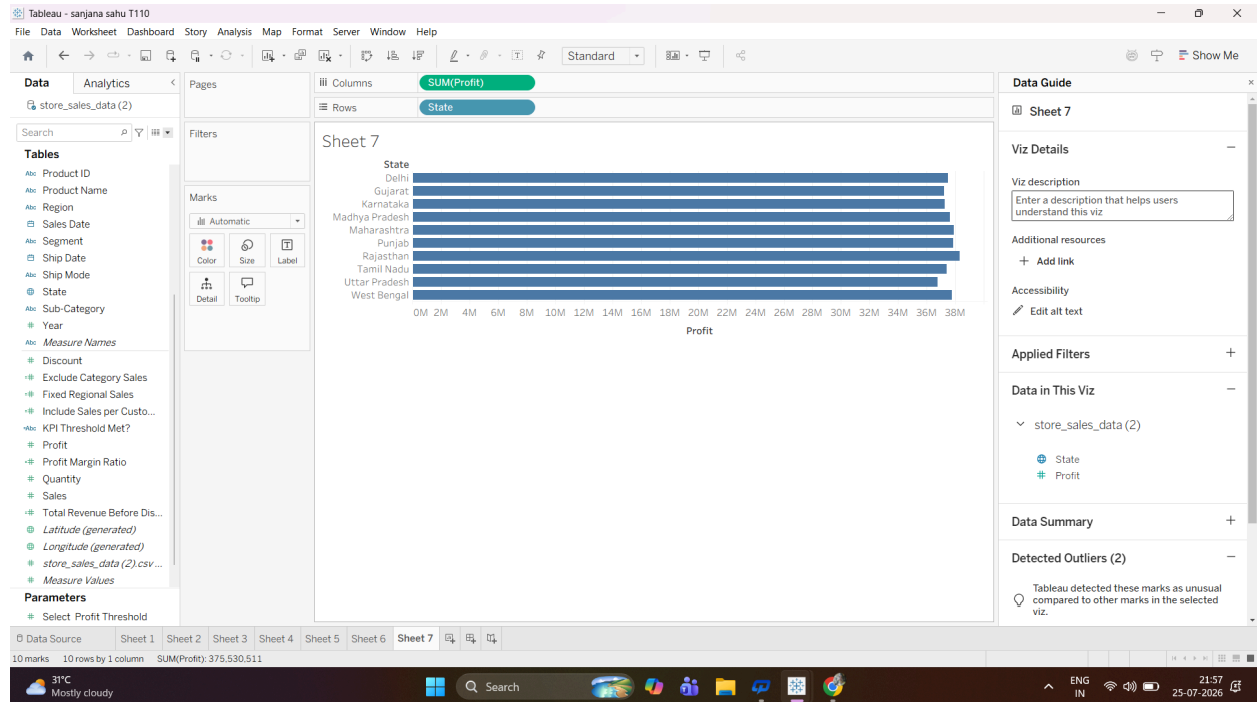
24 marks 24 rows by 1 column SUM of ATTR(Exclude Category Sales): 60,202,584,340

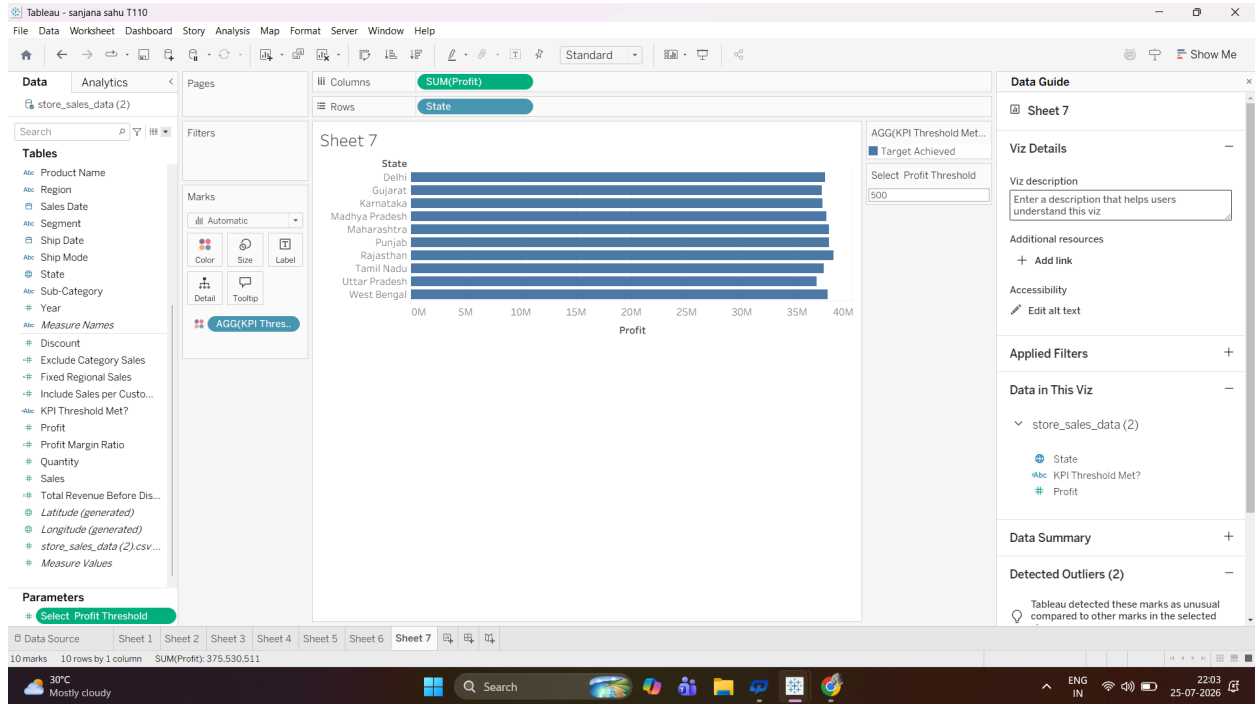
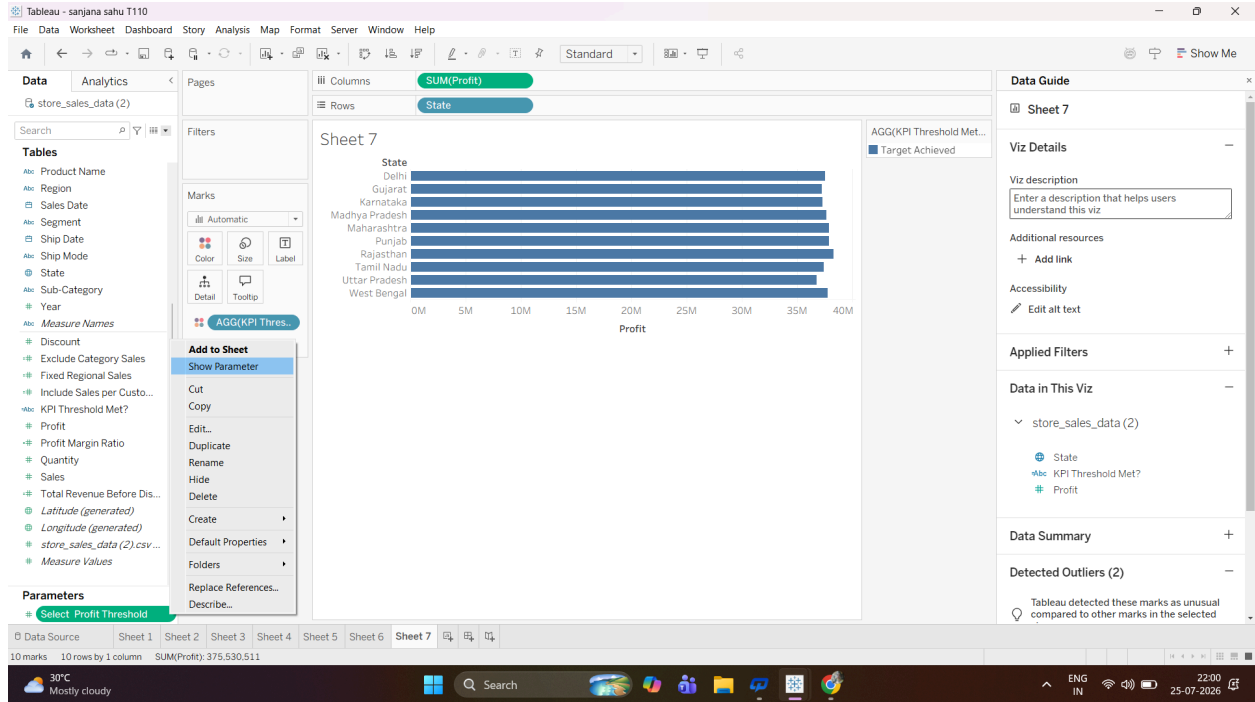
30°C Mostly cloudy

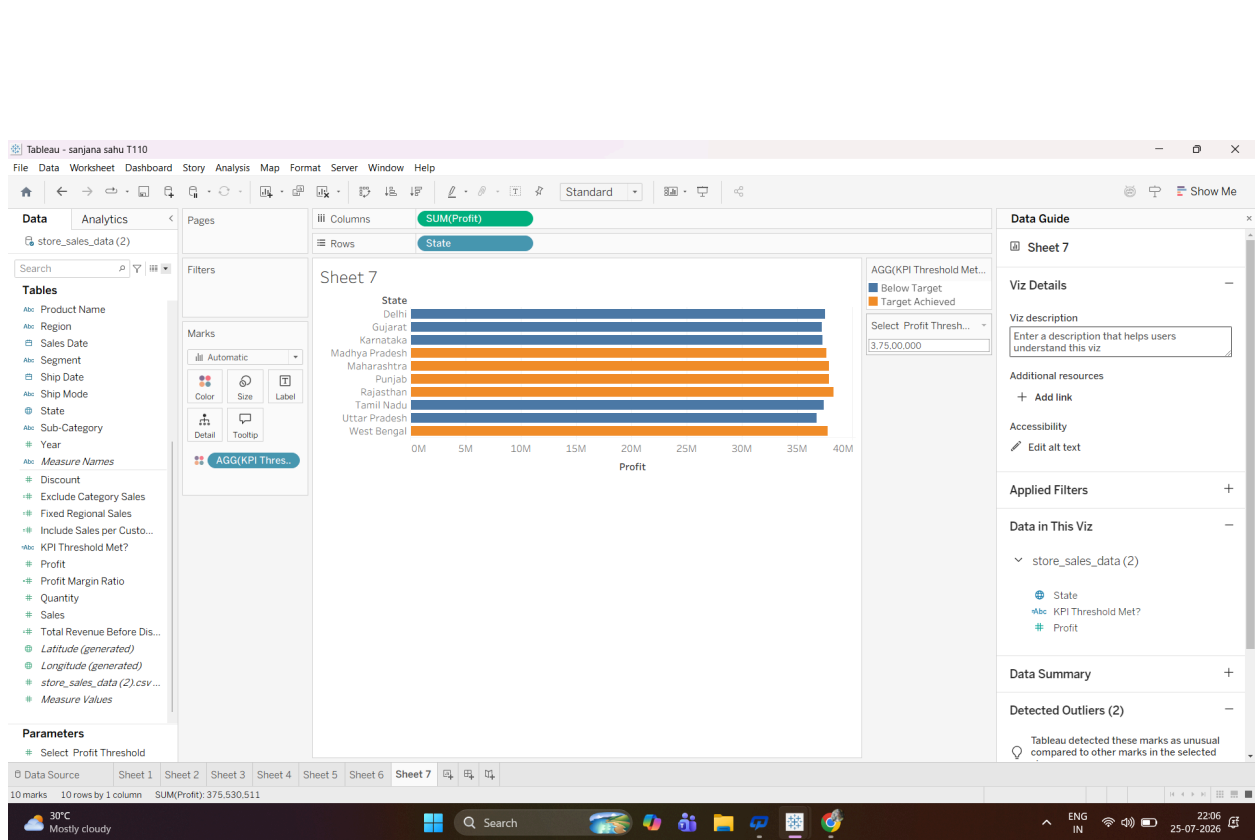
Search

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G]Apply calculations to fix data issues and enhance insights

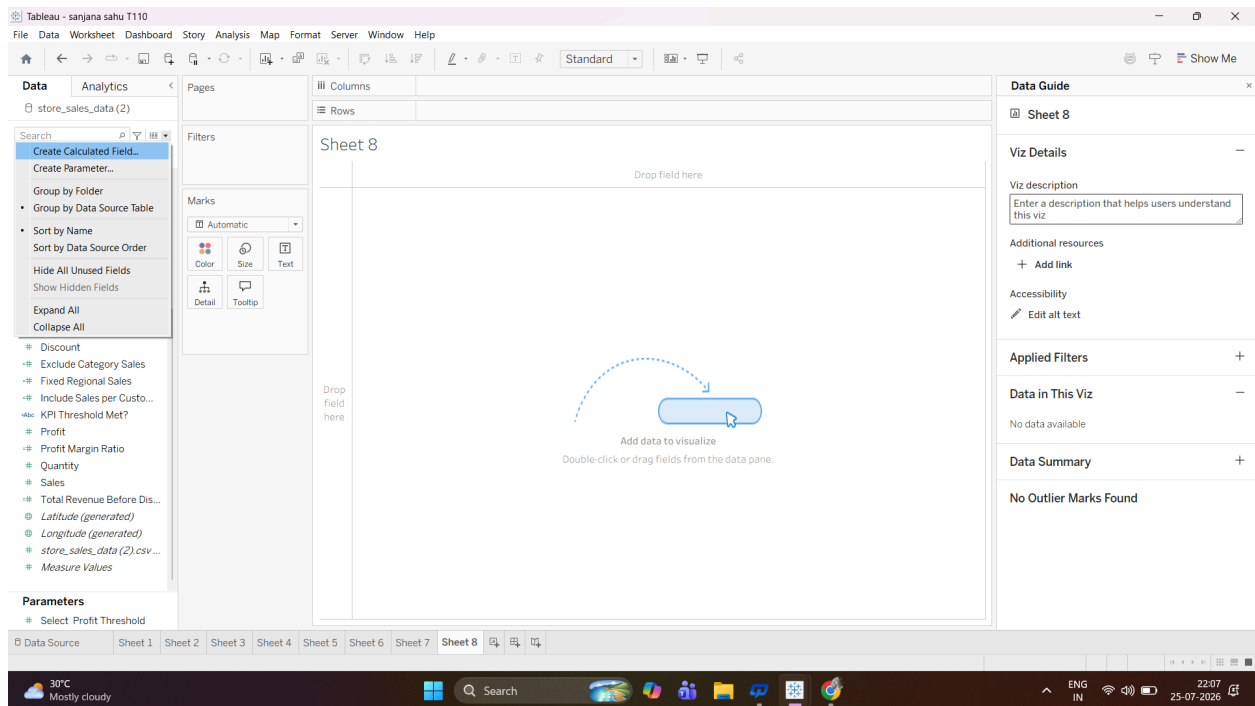


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File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Data Analytics

store_sales_data (2)

Search

Tables

- Product Name
- Region
- Sales Date
- Segment
- Ship Date
- Ship Mode
- State
- Sub-Category
- Year
- Measure Names
- Cleaned Discount**
- Discount
- Exclude Category Sales
- Fixed Regional Sales
- Include Sales per Custo...
- KPI Threshold Met?
- Profit
- Profit Margin Ratio
- Quantity
- Sales
- Total Revenue Before Dis...
- Latitude (generated)
- Longitude (generated)
- store_sales_data (2).csv...
- Measure Values

Parameters

- Select Profit Threshold

Columns

Rows

Filters

Drop field here

Marks

Automatic

Color Size Text

Detail Tooltip

Drop field here

Drop field here

Cleaned Discount

IFNULL([Discount],0)

The calculation is valid.

Apply OK Apply and Close

Add data to visualize

Double-click or drag fields from the data pane.

Data Guide

Sheet 8

Viz Details

Viz description

Enter a description that helps users understand this viz

Additional resources

+ Add link

Accessibility

Edit alt text

Applied Filters

+ Data in This Viz

- No data available

Data Summary

+ No Outlier Marks Found

Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8

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File Data Worksheet Dashboard Story Analysis Map Format Server Window Help

Standard

Data Analytics

store_sales_data (2)

Search

Create Calculated Field...

Create Parameter...

Group by Folder

Group by Data Source Table

Sort by Name

Sort by Data Source Order

Hide All Unused Fields

Show Hidden Fields

Expand All

Collapse All

- Cleaned Discount
- Discount
- Exclude Category Sales
- Fixed Regional Sales
- Include Sales per Custo...
- KPI Threshold Met?
- Profit
- Profit Margin Ratio
- Quantity
- Sales
- Total Revenue Before Dis...
- Latitude (generated)
- Longitude (generated)
- store_sales_data (2).csv...
- Measure Values

Parameters

- Select Profit Threshold

Columns

Rows

Filters

Drop field here

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Color Size Text

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Data Guide

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Viz description

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Edit alt text

Applied Filters

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- No data available

Data Summary

+ No Outlier Marks Found

Sheet 1 Sheet 2 Sheet 3 Sheet 4 Sheet 5 Sheet 6 Sheet 7 Sheet 8

30°C Mostly cloudy

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